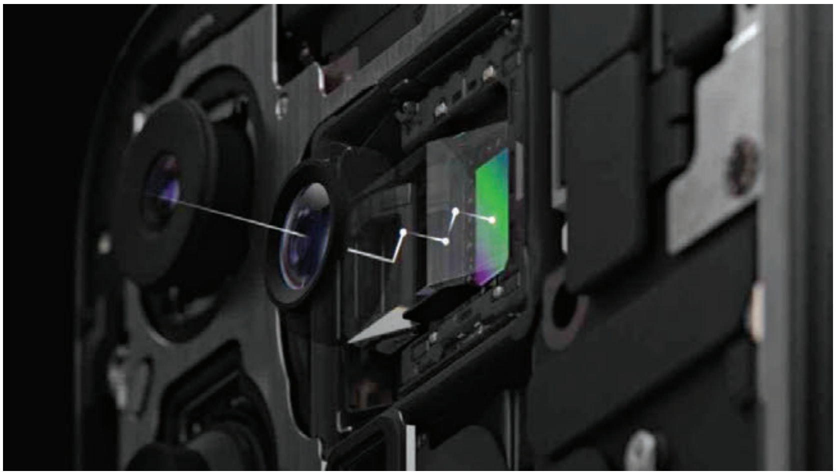
can buy, let's detail a few key points that you should be familiar with before you make your buying decision.

What is optical zoom? Do telephoto cameras on phones offer true optical zoom?

Almost all telephoto cameras on phones have a fixed focal length lens for their telephoto camera and not sliding lens elements, which means they support optical zoom at one focal length only. If your phone has 3X optical zoom, everything below 3x (2.9x, 2x, etc.) will be digital zoom or hybrid zoom. The only exception is perhaps the Sony Xperia 1 V, which integrates a movable periscope lens and can deliver optical zoom between 3.5 to 5.2x.

Also, the telephoto camera sensor is often smaller as compared to the primary sensor, and this allows manufacturers to get say "3X optical zoom" without using a lens that is 3x in focal length compared to the one used on the primary sensor.

The point is that most of the zooming action on phones is hybrid. It's the software and AI that sharpens and often reconstructs the zoom image at times adding pixels that don't exist, in order to make the end results appealing. That isn't to say that the optics don't help. Good telephoto hardware is needed for the software trickery to work convincingly. A few generations down the line, our phones are getting good



at zooming in and getting around the physics constraints.

Types of periscope cameras used in smartphones

Let's now discuss how the periscope telephoto camera works and address some of the popular periscope camera technologies and terminologies that you might run into when deciding between phones.

Periscope telephoto cameras – Periscope telephoto cameras on smartphones use prisms and mirrors and allow for a longer focal length or higher optical zoom without increasing the thickness of the phone. The idea is to give a longer path for light to travel within the lens assembly.

Floating periscope telephoto camera – In a floating periscope camera, the lens assembly isn't static, and parts of it can move. This helps with better stabilization as well as adds the ability to focus on both distant as well as nearby objects.

The Xiaomi 13 Pro, for instance, has a floating periscope camera. When the camera needs to capture images of far objects, the lens elements move closer together. This movement allows the lens to focus on distant subjects, providing clear and sharp images even at a considerable distance. On the other hand, when the camera needs to take close-up shots, such as capturing details of objects as close as 10 cm, the lens elements move away from each other.

Tetraprism telephoto camera – Apple's Tetraprims periscope camera used on the iPhone 15 Pro Max uses a series of four prisms to fold the light path within the camera. Apple also uses a 3D sensor-shift stabilization and autofocus module to negate handshakes which are prominent especially when you are all zoomed in.

Liquid lens periscope telephoto camera – Tecno used a liquid lens in its telephoto module, and the technology is expected to debut in Tecno flagship phones this year. This liquid lens adjusts its shape when voltage is applied, enabling dynamic focusing for close-up shots as near as 5cm. Just as in a floating lens assembly, this liquid lens will enable the telephoto module to focus on nearby objects and shoot macros.